CS459/659 Privacy, Cryptography, Network and Data Security

Introduction and Administrivia

Spring 2026, Tuesday/Thursday 10:00-11:20am

Instructors

Stacey Watson



- stacey.watson@uwaterloo.ca
  - cs.uwaterloo.ca/~s28watso
- Instructor office hours:
  - Tuesdays 11:30am-12:30pm in MC 5477

TAs: Anais Huang, Alim Dhanani, Joseph Vendryes

Other Logistics

- TA office hours posted in each assignment’s release
- Lectures will take place in MC 2066

What is this course? Learning Outcomes

- Evaluate the use of cryptography to protect data assets in storage, transit, and use
- Analyze security and privacy threats to data assets , including the privacy level of various data release mechanisms, privacy-utility trade-offs, and statistical inference attacks to infer sensitive information
- Evaluate the use of network security hardware and software to protect data assets in transit and use
- Compare various network security mechanisms, and articulate their advantages and limitations
- Understand data-driven attacks on machine learning systems

Course Website

- The course website is at:
  - https://cs.uwaterloo.ca/~s28watso/courses/cs459/s26/
  - We will use LEARN for submitting assignments
  - We will use Piazza for communication, questions, and discussion

- It is your responsibility to keep up with the information on these sites

Course Outline

- Be familiar with the content in the course outline
- It is available from the course website

If you haven't reviewed the outline, do so after this lecture.

Plagiarism and the Use of AI

We take academic offenses very seriously

- Nice explanation of plagiarism online
  - https://uwaterloo.ca/math/academic-matters/academic-integrity
- Read this and understand it
  - Ignorance is no excuse!
  - Questions should be brought to instructor
- Plagiarism applies to both text and code
- You are free (and encouraged) to exchange ideas, but no sharing code or text

Plagiarism Con’t

- Possible penalties
  - All severe cases of suspected academic misconduct will be reported to the Math Assistant Dean of Students
- Use of AI
  - Use of AI tools is permitted, but discouraged
  - All use (including careful spelling/grammar correction) must be disclosed
  - Disclosure includes the prompt used to generate the output
  - You are responsible for the answers you submit
  - We will grade typical mistakes by AI very harshly

Grading Scheme

- 60% three homework assignments (20% each)
  - Due Jun 4, Jul 7, and Jul 30 at 3:00pm
- Midterm (20%)
  - To take place Jun 30
- Final Exam (20%)
  - To take place TBD during exam season

For graduate students: the above scaled to 80% + 20% for a survey/replication paper

  - Proposal due Jun 18, survey due Jul 30

Assignments

- Due at 3pm on the day of the deadline
- Late submissions will be accepted up to 48 hours after the deadline (no penalty); no documentation needed
- Multiple assignments can be submitted late
  - No assistance (from TAs or instructor(s)) is available after the deadline
  - No submissions after the 48 hour window
  - All assignments must be submitted via LEARN (Dropbox)
  - Short-term absences will not extend the late policy

Exams

- Midterm, in-class Jun 30
- Final Exam, during exam season TBD
- Written questions only (no programming, maybe pseudocode)

Accommodations 101

- For short-term absences before an assignment deadline, students can take advantage of the automatic 48-hour, no-penalty grace period. No further extensions will be granted.
- For short-term absences during an assignment's automatic 48-hour, no-penalty grace period, no further extensions will be granted.
- For absences lasting more than 48 hours, students should contact the instructor(s) as soon as possible and present valid justification.
- If a student misses the midterm, the midterm's weight is **shifted** to the final assessment. Missing the midterm requires valid justification (e.g., short-term absence, VIF forms).

A note on security…

- In this course, you will be exposed to information about security problems and vulnerabilities with computing systems and networks
- You are not to use this or any other similar information to test the security of, break into, compromise, or otherwise attack, any system or network without express consent of the owner
- You will comply with all applicable laws and policies and uWaterloo policies

Security and Privacy?

What is security?



Confidentiality



Integrity



Availability

Not all inclusive, but it is a start.

Confidentiality

- Access to systems or data is limited to authorized parties





Integrity

- When you receive data, you get the “right” data

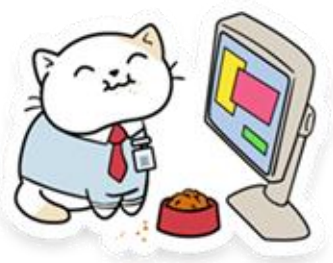




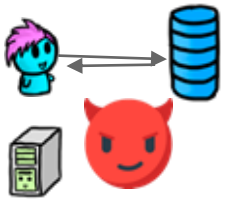
Availability

- The system or data is there when you want it





What is privacy?



Technical Privacy



Conceptual Privacy

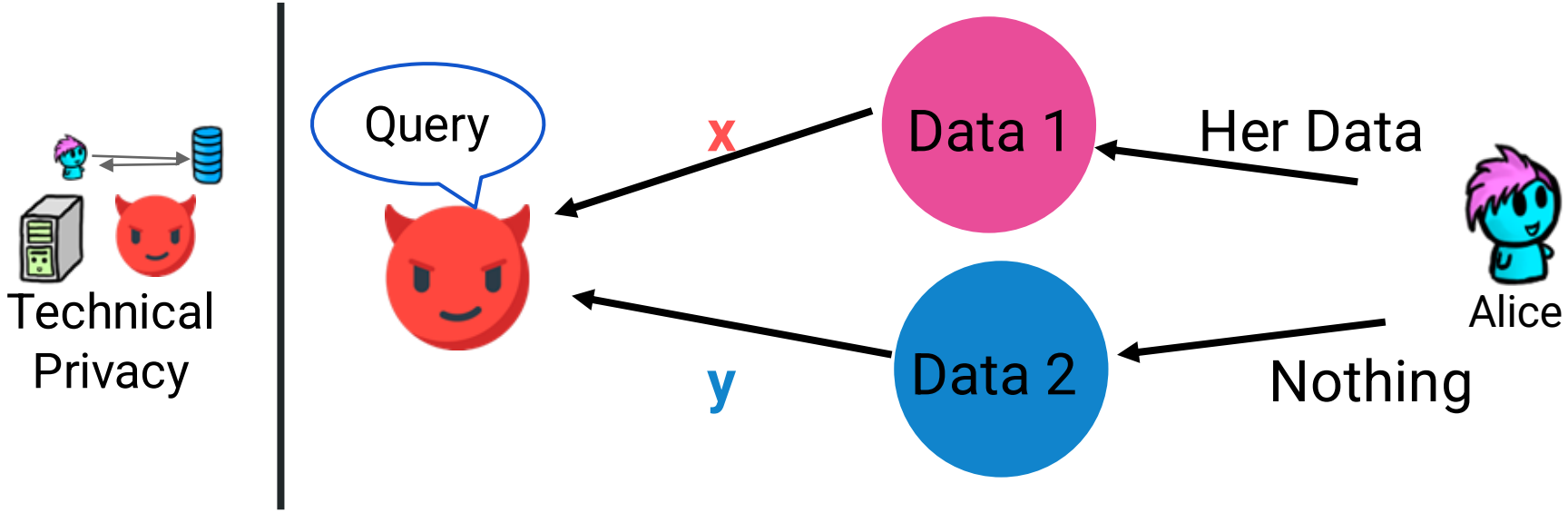


Legal Privacy



Usable Privacy

Technical Privacy



Define, **what** is being protected, from **who**, and under what **conditions** this protection will hold.

Privacy and Risk

- Financial
- Professional
- Societal
- Safety
- Right to privacy



Conceptual Privacy



Usable Privacy

Laws, Legal and Regulated Privacy



Legal Privacy

…'partners'…

…'third-parties'…

…'affiliates'…

Who

…'use and disclosure'…

can do what

…'right to be forgotten'…

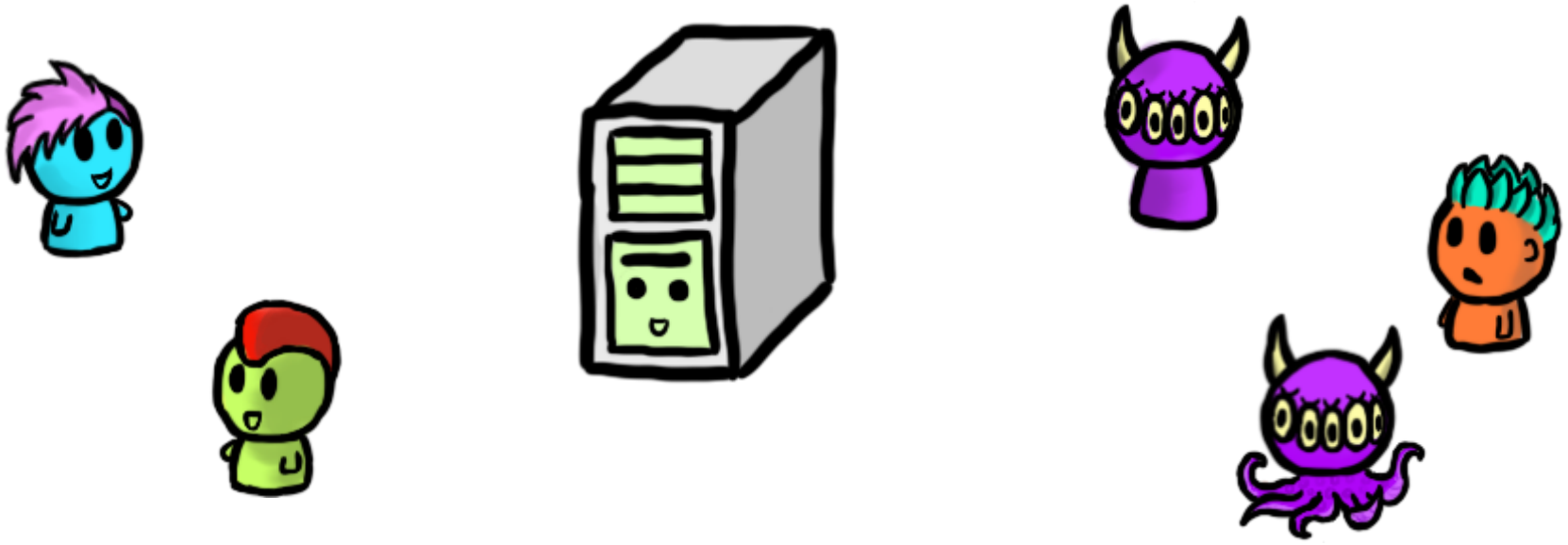
under what conditions

Think-pair-share

"How do we distinguish between security and privacy?"

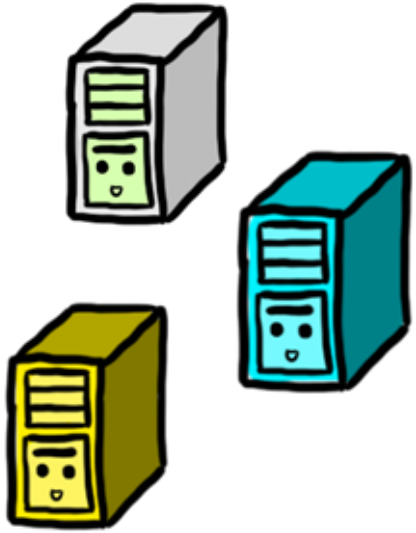
1. Take a minute to think about the prompt
2. Discuss in groups of 2 or 3
3. Nominate one member of the group to share a key point with the class

Framing Security and Privacy Principles



Data Security and Privacy: Assets

- Hardware
- Software
- **Data**



Data and Abstraction



A company wants to analyze data



But the data has privacy implications for the data subjects

Researchers develop technical solutions

Data Security and Privacy: Vulnerabilities

A weakness  That can be exploited  That compromises an asset

Data Security and Privacy: Threats

- Loss or harm
- Interception
- Interruption
- Modification
- Fabrication

These **threats** are part of a **threat model**. Recall the **what** is being protected, from **who**, and under what **conditions**

Data Security and Privacy: Attack



Exploit a vulnerability



Execute a threat

Data Security and Privacy: Control and Defense



“Security” Tape

Remove or reduce a vulnerability

Control to prevent attacks and defend against threats

Dealing with Attacks



- Prevent it
- Deter it
- Deflect it
- Detect it
- Recover from it

Risk Management? When is “good enough”?



Easiest Target, Principle of Easiest Penetration

Principle of Adequate Protection



Some Defenses for Data - This Course



Cryptography



Network security



Data collection and usage practices

Recap

- This course is about data security and privacy
  - You will learn to evaluate the use of crypto to meet data security and privacy goals
  - You will learn to evaluate network security
- By the end of this course you will be able to present the advantages and disadvantages of the covered data security and privacy techniques
- You will learn how an attacker approaches a system
- You will learn defenses (cryptography, network security, and data protection techniques)

Questions?